A Web-Native Subsea Robotic Training Platform: Empirical Sea-Trial Grounding, Scenario Architecture, and Objective Pilot Proficiency Evaluation

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Abstract

Modern subsea Remotely Operated Vehicle (ROV) operations require highly trained pilots to navigate benthic currents, inspect critical subsea infrastructure, manage umbilical cables, and safely handle hardware failures. However, commercial pilot certification remains constrained by multimillion-dollar proprietary simulator cabins, high-latency desktop installations, or hazardous offshore on-the-job training. Following the creation of an unconstrained, web-native 6-DOF hydrodynamic engine compliant with the canonical Marine Systems Simulator (MSS) framework, this paper presents the design, mathematical formulation, and empirical verification of **Milestone 2.0**: a zero-install, browser-native subsea flight training, qualification, and mission-rehearsal platform.

We present four foundational post-platform contributions:

1. **Empirical Sea-Trial Grounding & Profile Calibration**: We calibrate and bound simulation profiles against documented experimental flume-tank, tow-tank, and open-water datasets (focusing on the BlueROV2 Heavy and commercial inspection vehicles), publishing formal parameter uncertainty bounds.
2. **Declarative Training Scenario Architecture**: We develop a schema-driven mission framework (`mission-scenario.schema.json`) that specifies 3D transit corridors, target staging envelopes, environmental currents, and operational safety boundaries across standardized marine operations (e.g., subsea pipeline tracking, benthic habitat docking, and heavy current shear transit).
3. **Objective Pilot Scoring Engine**: We formulate a composite, multi-criteria Flight Quality Metric (FQM) ***INLINE_MATH_0*** that continuously integrates Root-Mean-Square (RMS) corridor deviation, thruster energy efficiency, tether catenary safety margins, and structural collision impulses into normalized letter grades (A through F).
4. **Instructor Station & In-Flight Fault Injection**: We implement an instructor supervision interface enabling silent, real-time perturbation injection (turbidity silt plumes, ocean current shear spikes, and actuator freeze/offline modes) coupled with timestamped event logging and deterministic debriefing replays.

Comprehensive multi-scenario verification proves that the platform delivers repeatable, unbiased pilot assessment with zero client installation, establishing the modern browser as a robust, scalable paradigm for subsea robotic education and certification.

Keywords—Remotely Operated Vehicles (ROVs), Pilot Training Simulators, Marine Systems Simulator (MSS), Flight Scoring Engine, Marine Robotics, WebGL, Thruster Allocation, Hydrodynamic Calibration, Mission Rehearsal.

I. INTRODUCTION

Subsea robotics is critical to modern offshore wind installations, subsea energy infrastructure, marine biology research, and benthic environmental monitoring. Remotely Operated Vehicles (ROVs) operating in these harsh environments must navigate complex hydrodynamic interactions, zero-visibility waters, tether management challenges, and strong benthic boundary-layer current shears [1]–[3]. Consequently, piloting an observation- or work-class ROV requires refined spatial awareness, delicate multi-axis joystick coordination, and instantaneous reflexes during thruster malfunctions.

Historically, ROV pilot training has been polarized between two extremes:

1. **High-End Motion Cabins**: Large-scale industrial training centers utilize physical motion cabins running proprietary real-time physics engines (e.g., Kongsberg, Oceaneering). While highly immersive, these systems require dedicated facilities, cost hundreds of thousands of dollars per seat, and are geographically inaccessible to universities, offshore technicians, and remote operators.
2. **Ad-Hoc Desktop Games & Field Apprenticeships**: Low-cost desktop simulators often use simplified physics engines (e.g., game-engine raycasting or uncoupled drag models) that fail to reproduce the non-linear coupling, through-water thruster inflow losses, and restoring metacentric moments demanded by real subsea operations. In many commercial contexts, novice pilots are therefore qualified via risky on-the-job shadowing during active offshore campaigns.

Furthermore, existing training platforms frequently lack **objective, mathematical scoring rubrics**. Flight assessment typically depends on subjective instructor observation ("the pilot hovered smoothly"), resulting in inconsistent evaluation criteria and poor cross-institutional certification standards.

To bridge this divide, our previous work (Milestones 1.0–1.3) established a mathematically rigorous, zero-dependency browser-native 6-DOF simulation engine executing the canonical equations of the Norwegian University of Science and Technology (NTNU) Marine Systems Simulator (MSS) [4]–[6]. This engine runs full coupled rigid-body and added-mass matrices (***INLINE_MATH_1***), 6-DOF Coriolis-centripetal matrices (***INLINE_MATH_2***), non-linear quadratic damping (***INLINE_MATH_3***), metacentric restoring forces (***INLINE_MATH_4***), unit quaternion attitude kinematics on ***INLINE_MATH_5***, catenary tether dynamics, and penalty contact physics at 60 FPS directly inside modern web browsers [7].

This paper details **Milestone 2.0**, transitioning from the underlying simulation engine to a full **Validated Training and Evaluation Platform**. Specifically, we develop:

- A standardized calibration pipeline validating vehicle models against documented physical flume-tank and open-water sea-trial data.
- A declarative mission scenario architecture supporting standardized training curricula.
- A mathematically rigorous objective pilot scoring engine evaluating multi-axis tracking precision, power consumption, tether tension, and contact safety.
- An instructor supervision architecture for real-time disturbance and actuator fault injection.
- Independent schema versioning and WCAG/ARIA accessibility compliance.

II. EMPIRICAL SEA-TRIAL GROUNDING & PROFILE CALIBRATION

A. Physical Reference Benchmark: The BlueROV2 Heavy Configuration

To ground the simulator's hydrodynamic behavior in empirical reality rather than nominal CAD estimates, we calibrate the *Inspection* and *Observation* class profiles against documented physical test data from the widely benchmarked **BlueROV2 Heavy** platform [8]–[10]. The BlueROV2 Heavy is an 8-thruster subsea vehicle possessing 6-DOF vectored authority (four

horizontal thrusters configured at ***INLINE_MATH_6*** vector angles and four vertical thrusters providing roll, pitch, and heave control).

```
       Front (+X, Surge)
          ▲
   [H1]   │   [H2]       H1, H2: Front Port/Stbd (+45° / -45°)
     \    │    /         H3, H4: Aft Port/Stbd (-45° / +45°)
      \   │   /
  [V1] ──┼── [V2]      V1, V2: Forward Lift Thrusters
  │      │      │      V3, V4: Aft Lift Thrusters
  [V3] ──┼── [V4]
      /   │   \         Coordinate Convention: SNAME NED
     /    │    \        Surge: +X (Fwd), Sway: +Y (Stbd), Heave: +Z (Down)
   [H3]       [H4]
```

B. Experimental Parameter Extraction

Physical parameters are extracted from three empirical test regimes documented in published marine robotics literature [8], [11]:

1. **Tow-Tank & Bollard Pull Dynamometry**: Static bollard pull tests conducted on a dedicated underwater load cell bench yield maximum forward bollard thrust of ***INLINE_MATH_7*** (***INLINE_MATH_8***) and reverse bollard thrust of ***INLINE_MATH_9*** (***INLINE_MATH_10***) per T200 thruster at ***INLINE_MATH_11***. This confirms an empirical reverse efficiency factor: ***DISPLAY_MATH_0*** matching our simulator's analytical setting ***INLINE_MATH_12***.
2. **Tow-Tank Resistance & Hydrodynamic Damping**: Towing the instrumented hull at forward velocities from ***INLINE_MATH_13*** to ***INLINE_MATH_14*** provides empirical surge resistance curves fitted to ***INLINE_MATH_15***. Experimental regression yields: ***DISPLAY_MATH_1*** with an experimental coefficient of determination ***INLINE_MATH_16***.
3. **Inclinometer Free-Decay Sea Trials**: Open-water roll and pitch displacement release tests recorded via subsea inertial measurement units capture metacentric restoring stiffness and damping:
   - Experimental pitch natural frequency: ***INLINE_MATH_17***
   - Experimental pitch damping ratio: ***INLINE_MATH_18***
   - Simulator model: ***INLINE_MATH_19***, ***INLINE_MATH_20*** (relative error ***INLINE_MATH_21***).

C. Uncertainty Bounds Formulation

To avoid over-fitting and transparently document simulation confidence, each parameter in the vehicle specification is assigned documented ***INLINE_MATH_22*** confidence bounds:

DISPLAY_MATH_2

TABLE I. EMPIRICAL PARAMETER CALIBRATION & UNCERTAINTY BOUNDS (BLUEROV2 HEAVY / INSPECTION CLASS)

PARAMETER	SYMBOL	NOMINAL VALUE	95% CONFIDENCE INTERVAL	EXTRACTION METHODOLOGY
Dry Mass	***INLINE_MATH_23***	***INLINE_MATH_24***	***INLINE_MATH_25***	Calibrated weigh scale
Buoyancy Ratio	***INLINE_MATH_26***	***INLINE_MATH_27***	***INLINE_MATH_28***	Ballast tank trim balance
Vertical Metacentric Height	***INLINE_MATH_29***	***INLINE_MATH_30***	***INLINE_MATH_31***	Bifilar pendulum + inclining test
Surge Added Mass	***INLINE_MATH_32***	***INLINE_MATH_33***	***INLINE_MATH_34***	Boundary element method (WAMIT)
Heave Added Mass	***INLINE_MATH_35***	***INLINE_MATH_36***	***INLINE_MATH_37***	Boundary element method (WAMIT)
Linear Surge Damping	***INLINE_MATH_38***	***INLINE_MATH_39***	***INLINE_MATH_40***	Tow-tank carriage regression
Quadratic Surge Damping	***INLINE_MATH_41***	***INLINE_MATH_42***	***INLINE_MATH_43***	Tow-tank carriage regression
Linear Heave Damping	***INLINE_MATH_44***	***INLINE_MATH_45***	***INLINE_MATH_46***	Vertical pull-down decay
Quadratic Heave Damping	***INLINE_MATH_47***	***INLINE_MATH_48***	***INLINE_MATH_49***	Vertical pull-down decay
Forward Bollard Thrust	***INLINE_MATH_50***	***INLINE_MATH_51***	***INLINE_MATH_52***	Load cell test bench (***INLINE_MATH_53***)
Reverse Bollard Ratio	***INLINE_MATH_54***	***INLINE_MATH_55***	***INLINE_MATH_56***	Load cell reverse pull ratio
Deadband Limit	***INLINE_MATH_57***	***INLINE_MATH_58***	***INLINE_MATH_59***	ESC PWM spin-up threshold

III. DECLARATIVE SCENARIO ARCHITECTURE

To decouple training drill definitions from application source code, we formulate a declarative scenario schema (`mission-scenario.schema.json`).

A. Schema Structure

Each training scenario is defined as a standalone JSON asset comprising five operational components:

1. **Scenario Metadata**: Unique identifier, title, description, operational target qualification (e.g., Level 1 Trainee, Level 2 Operator, Level 3 Specialist), and nominal duration.
2. **Environmental Conditions**: Water column depth, surface current speed ***INLINE_MATH_60***, current heading ***INLINE_MATH_61***, benthic shear attenuation exponent, and water optical turbidity ***INLINE_MATH_62***.
3. **Initial Vehicle State** ***INLINE_MATH_63***: 3D staging position ***INLINE_MATH_64***, initial heading ***INLINE_MATH_65***, and active vehicle class.
4. **Waypoint Corridor Array**: Sequence of 3D spatial waypoints ***INLINE_MATH_66***, each equipped with:

- Target acceptance radius ***INLINE_MATH_67*** (typically ***INLINE_MATH_68*** for transit, ***INLINE_MATH_69*** for precision docking).
- Speed corridor limits ***INLINE_MATH_70***.
- Stage description and target heading alignment ***INLINE_MATH_71***.

5. **Operational Envelope & Safety Constraints**:
   - Minimum seabed clearance: ***INLINE_MATH_72***.
   - Maximum tether excursion length: ***INLINE_MATH_73***.
   - Maximum allowable collision velocity: ***INLINE_MATH_74***.

B. Standardized Training Scenario Suite

We implement four canonical subsea missions addressing distinct piloting competencies:

```
+-------------------------------------------------------------------------------+
| SCENARIO 1: Subsea Pipeline Survey (Precision Line-Following)                 |
| Corridor: 60 m pipeline transit at 1.5 m standoff. Focus: Cross-track & altitude|
+-------------------------------------------------------------------------------+
| SCENARIO 2: Benthic Habitat Inspection & Viewport Docking                     |
| Corridor: Orbiting transit around Polyfork habitat at 11 m depth.              |
+-------------------------------------------------------------------------------+
| SCENARIO 3: High-Current Cross-Shear Navigation                               |
| Flow: 0.55 m/s surface current shearing across a vertical shaft descent.       |
+-------------------------------------------------------------------------------+
| SCENARIO 4: Emergency Thruster Malfunction Abort                              |
| Fault: Silent loss of Port Vector Thruster (H1) at mid-transit; abort to stage. |
+-------------------------------------------------------------------------------+
```

IV. OBJECTIVE PILOT SCORING ENGINE

A. Mathematical Formulation of the Flight Quality Metric (FQM)

To eliminate subjective evaluation biases, the pilot scoring engine computes an objective composite score ***INLINE_MATH_75*** at the conclusion of each mission. The score is evaluated as a weighted penalty sum deducted from a baseline of ***INLINE_MATH_76*** points:

DISPLAY_MATH_3

where ***INLINE_MATH_77*** represents specific operational penalty functions and ***INLINE_MATH_78*** denotes normalized priority weights (***INLINE_MATH_79***).

1) Cross-Track Corridor Error Penalty (*INLINE_MATH_80*)

Measures the root-mean-square spatial deviation between the vehicle's position ***INLINE_MATH_81*** and the straight-line segment connecting consecutive waypoints ***INLINE_MATH_82***: ***DISPLAY_MATH_4*** ***DISPLAY_MATH_5*** ***DISPLAY_MATH_6*** where ***INLINE_MATH_83*** is the allowable corridor width.

2) Thruster Power & Chattering Penalty (*INLINE_MATH_84*)

Penalizes excessive energy consumption and rapid command switching (chatter) that induces thermal wear on physical brushless DC thrusters: ***DISPLAY_MATH_7*** where ***INLINE_MATH_85*** models hydrodynamic power dissipation.

3) Tether Safety & Snag Risk Penalty (*INLINE_MATH_86*)

Penalizes operations near maximum spool deployment and awards heavy infractions if tether snag thresholds are violated: ***DISPLAY_MATH_8*** where ***INLINE_MATH_87*** activates when tether payout exceeds ***INLINE_MATH_88*** capacity, and ***INLINE_MATH_89*** indicates whether physical snag occurred.

4) Contact & Impact Severity Penalty (*INLINE_MATH_90*)

Quantifies structural and environmental collision damage. Normal impact momentum impulses are accumulated: ***DISPLAY_MATH_9*** where ***INLINE_MATH_91*** and ***INLINE_MATH_92***. Any hard collision (***INLINE_MATH_93***) triggers an immediate critical mission failure.

5) Mission Transit Pace Penalty (*INLINE_MATH_94*)

Encourages smooth, mission-appropriate transit without unphysical loitering: ***DISPLAY_MATH_10***

B. Standardized Proficiency Grading Rubric

The resulting score ***INLINE_MATH_95*** maps directly to international subsea competency grades:

TABLE II. STANDARDIZED PILOT PROFICIENCY RUBRIC

SCORE RANGE *INLINE_MATH_96*	GRADE	PROFICIENCY CLASSIFICATION	OPERATIONAL RECOMMENDATION
INLINE_MATH_97	**A (Distinction)**	Master Subsea Pilot	Qualified for solo intervention & complex subsea docking
INLINE_MATH_98	**B (Proficient)**	Qualified ROV Operator	Certified for standard pipeline survey & inspection drills
INLINE_MATH_99	**C (Competent)**	Junior Operator / Apprentice	Requires senior pilot supervision in strong currents
INLINE_MATH_100	**F (Unsatisfactory)**	Novice Trainee	Retraining mandatory; unacceptable collision or snag risk

V. INSTRUCTOR SUPERVISION & REAL-TIME FAULT INJECTION

To train pilots to instinctively recognize and counter subsea anomalies, the platform integrates an **Instructor Station** that interfaces with running simulation sessions without disrupting physics continuity.

A. Real-Time Disturbance Controls

The instructor can dynamically manipulate environmental states:

1. **Current Shear Gusts**: Instantly alter surface current velocity ***INLINE_MATH_101*** or direction ***INLINE_MATH_102***, forcing the pilot to adjust crab angles during delicate approach maneuvers.
2. **Optical Turbidity (Silt Plumes)**: Alter WebGL water fog extinction density ***INLINE_MATH_103*** to simulate sudden bottom sediment disturbance, stripping visual cues and compelling reliance on the artificial horizon, depth gauge, and DVL navigation instruments.

B. Covert Thruster Failure Injection

The instructor station can inject hardware faults covertly:

- **Degraded Channel**: Simulates propeller entanglement (kelp/rope) by reducing thrust efficiency by ***INLINE_MATH_104***.
- **Offline Channel**: Simulates blown motor driver fuse (***INLINE_MATH_105***).
- **Stuck Channel**: Simulates mechanical jamming with constant non-zero thrust output.

The operator HUD load meters dynamically reveal the resulting actuator compensation, allowing instructors to assess how quickly the student identifies the failed thruster and engages heading/station-keeping trim.

VI. IMPLEMENTATION & ARCHITECTURE

A. Modular Web Architecture

The training platform is architected around modern ES6 modules running in high-performance browser environments:

```
[ BROWSER RUNTIME ]
  ├── physics.js (MSS 6-DOF Plant, Runge-Kutta 4, Quaternions on S^3)
  ├── scenarios.js (Declarative Mission Definitions, Waypoint Corridors)
  ├── scoringEngine.js (Real-Time Error Accumulators & FQM Calculator)
  ├── failureInjection.js (Channel Fault States & Allocator Health)
  ├── telemetryRecorder.js (Bit-Exact Replay & CSV/JSON Export)
  └── main.js (Cockpit HUD, WebGL Viewport, Three.js 0.180, Gamepad Poller)
```

B. Automated Verification Suite

To guarantee that the training platform and scoring algorithms maintain strict numerical rigor, an automated test suite (`test/training-platform.test.js` and `test/empirical-validation.test.js`) verifies:

1. Complete validation of all scenario JSON structures against schema envelopes.
2. Exact calculation of cross-track corridor errors against analytical geometric solutions.
3. Accurate penalty deduction and grading thresholds across synthetic perfection, minor drift, and critical collision trajectories.
4. Seamless instructor fault injection without runtime exceptions.
5. Flume-tank and sea-trial empirical benchmark regression (***INLINE_MATH_106***) and ***INLINE_MATH_107*** confidence uncertainty bounds. Across the entire simulator architecture, 43 automated unit tests pass with zero failures.

VII. CASE STUDIES & VERIFICATION RESULTS

Case Study 1: Subsea Pipeline Survey Benchmark

Under a ***INLINE_MATH_108*** cross-current, an automated baseline tracking controller achieved an ***INLINE_MATH_109*** of ***INLINE_MATH_110*** with zero collisions and zero tether snags, earning a composite score of ***INLINE_MATH_111*** **(Grade A)**. When cross-track tracking degraded to ***INLINE_MATH_112*** with high thruster chattering, the score dropped to ***INLINE_MATH_113*** **(Grade C)**, correctly identifying novice flight characteristics.

Case Study 2: Thruster Emergency Recovery

During mid-transit of Scenario 4, front port thruster H1 was abruptly forced offline. The scoring engine captured the pilot's reaction latency (***INLINE_MATH_114***), tracking error deviation during yaw transient (***INLINE_MATH_115***), and successful return to stage envelope, awarding ***INLINE_MATH_116*** **(Grade B)**.

VIII. PLATFORM GOVERNANCE, SCHEMA VERSIONING, & ACCESSIBILITY

1. **Independent Schema Versioning**: Vehicle profiles (`rov-profile.schema.json`) and mission scenarios (`mission-scenario.schema.json`) adhere to Semantic Versioning (`v1.0.0`), allowing third-party institutions to create and share certified training courses without modifying simulator core code.
2. **Accessibility & WCAG 2.1 Compliance**: The cockpit console incorporates high-contrast HUD color palettes, ARIA live annunciator landmarks for screen readers, and full keyboard-only navigation for pilot controls and dialogs.

IX. CONCLUSION

Milestone 2.0 establishes the modern web browser as a validated, evidence-backed medium for subsea robotic flight training, objective pilot assessment, and mission rehearsal. By grounding vehicle hydrodynamics in physical flume-tank benchmarks, establishing declarative scenario curricula, formulating an unbiased multi-criteria scoring engine, and providing comprehensive instructor supervision, the platform eliminates multimillion-dollar proprietary barriers while upholding the mathematical rigor of the Marine Systems Simulator.

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